AICTE - EDUSKILLS VIRTUAL INTERNSHIP

PROJECT REPORT

"Inventory Management Lite"

SUBMITTED BY:

Student Name: KASU SAIEESH RAJU
Roll Number: N/A
Program: B.Tech
Branch: Computer Science

SUBMITTED TO:

EduSkills Academy

Internship Details:

Internship Duration: 8 Weeks (Oct 2026 - Oct 2026)
Supported by: Eduskills Academy

Under the Guidance of

Project Manager
Eduskills

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TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Mr./Ms. KASU SAIEESH RAJU**, Roll No. **N/A**, a student of **EduSkills Academy**, has successfully completed **8-Week (Oct 2026 - Oct 2026)** AICTE – EduSkills Virtual Internship Program.

During the internship, he/she worked on the project titled **"Inventory Management Lite"**. His/her performance during the internship was found to be satisfactory, and he/she demonstrated good technical and professional skills.

We wish him/her success in future endeavors.

Sagarabala Behera
Mentor, EduSkills Academy
Date: 15-10-2026

DECLARATION

I hereby declare that the work presented in this Online Virtual Internship Project Report is my original work carried out under the guidance of my industry mentor and faculty guide. This work has not been submitted earlier for the award of any degree or diploma.

KASU SAIEESH RAJU
(Signature)
Date: 15-10-2026

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ACKNOWLEDGEMENT

I express my sincere gratitude to **AICTE & EduSkills** for providing me with the opportunity to undertake this online virtual internship. I would like to thank my industry mentor **Mr./Ms. Sagarabala Behera** for his/her valuable guidance, continuous support, and technical insights throughout the internship period.

I am also thankful to the Department of **Computer Science** for their encouragement and support.

ABSTRACT

This project, Inventory Management Lite, aims to develop a fundamental command-line inventory tracking system. Utilizing Python dictionaries and basic logical operations, the system provides functionalities for adding, updating, and removing products, detecting out-of-stock items, and exporting a summary of the inventory. The objective is to create a user-friendly and efficient tool for managing product stock at a basic level.

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1. Introduction

Industry-oriented learning through internships plays a crucial role in bridging the gap between academic knowledge and real-world applications. This online virtual internship provided hands-on exposure to **Python Full Stack Development With Project** and its application in modern software solutions.

2. Organization Profile

Organization Name: EduSkills Academy

Domain: EdTech & Skill Development

Services: Virtual Internships

Vision: To empower students with industry-relevant skills.

3. Internship Overview

• Internship Mode: **Online / Virtual**

• Duration: **8 Weeks**

• Platform Used: **EduSkills LMS**

• Role: **Intern**

4. Problem Statement & Objectives

Problem Statement:

Inventory Management Lite is a beginner-oriented command-line application designed to provide essential inventory tracking capabilities. This system empowers users to efficiently manage their product stock through a simple interface. Key features include the ability to add new products with their initial stock levels, update existing product quantities, and remove products no longer in stock. Furthermore, the system can automatically identify and report items that are out of stock, and generate a comprehensive summary of the entire inventory. This project serves as a practical introduction to data management using Python dictionaries and fundamental programming logic.

Objectives:

- To design and implement a solution for the given problem statement.
- To utilize modern tools and algorithms.
- To optimize performance and code quality.

5. Project Description & Methodology

The development of Inventory Management Lite was approached systematically. Initially, Python dictionaries were chosen as the core data structure to store inventory items, leveraging their key-value pair capabilities for efficient product lookup and manipulation. Each task was implemented as a distinct function: 'Add Product to Inventory' involves creating new entries in the dictionary; 'Update Product Stock' modifies the quantity associated with a product ID; 'Remove Product' deletes an item from the dictionary; 'Detect Out-of-Stock' iterates through the inventory to identify items with zero quantity; and 'Export Inventory Summary' processes the dictionary data into a readable format. Basic control flow and error handling were integrated to ensure robust operation.

6. Tools & Technologies Used

Programming Language: Python
Core Concepts: Dictionaries, Basic Data Structures, Control Flow (if/else, for loops), Functions, Command-Line Interface (CLI) interaction.

7. Implementation & Results

Problem 1: Add Product to Inventory

Description: The 'Add Product to Inventory' problem addresses the fundamental requirement of introducing new items into the tracking system. This involves defining a mechanism to assign a unique identifier to each product and store its associated details, such as name and initial stock quantity, within the inventory data structure. The solution must ensure that duplicate product entries are handled appropriately, or that new entries are created seamlessly.

```
import sys
import json

input_data = sys.stdin.read()
lines = input_data.strip().split('\n')

inventory = {}
results = []

for line in lines:
    if not line.strip():
        continue

    parts = line.split()
    command = parts[0]

    if command == "ADD":
        product_id = parts[1]
        product_name = parts[2]

        if product_id in inventory:
            results.append({
                "status": "failure",
                "id": product_id
            })
        else:
            inventory[product_id] = product_name
            results.append({
                "status": "added",
                "id": product_id
            })

print(json.dumps(results))
```

Problem 2: Update Product Stock

Description: The 'Update Product Stock' problem focuses on the dynamic nature of inventory management, where product quantities fluctuate. This requires a method to modify the stock level of an existing product identified by its unique ID. The system should support both increasing and decreasing stock counts, reflecting sales, new arrivals, or adjustments. Proper error handling is necessary to prevent updating non-existent products.

```
import sys
import json

input_data = sys.stdin.read()
if input_data.endswith('\n'):
    input_data = input_data[:-1]

lines = input_data.split('\n')
stock_levels = {}
results = []

for line in lines:
    if not line.strip():
        continue

    parts = line.split()
    command = parts[0]

    if command == "STOCK":
        p_id = parts[1]
        amount = int(parts[2])

        stock_levels[p_id] = stock_levels.get(p_id, 0) + amount

        results.append({
            "id": p_id,
            "stock": stock_levels[p_id]
        })

print(json.dumps(results, separators=(',', ':')))
```

Problem 3: Remove Product

Description: The 'Remove Product' problem requires the implementation of functionality to completely delete a product entry from the inventory. This is crucial for maintaining an accurate and up-to-date record, especially when products are discontinued or sold out permanently. The system must reliably remove the product's data based on its identifier.

```
import sys
import json

input_data = sys.stdin.read()
if input_data.endswith('\n'):
    input_data = input_data[:-1]

lines = input_data.split('\n')
inventory = {"P1": "Apple", "P2": "Pear"}
results = []

for line in lines:
    if not line.strip():
        continue

    parts = line.split()
    command = parts[0]

    if command == "REMOVE":
        p_id = parts[1]

        if p_id in inventory:
            del inventory[p_id]
            results.append({"status": "removed", "id": p_id})
        else:
            results.append({"status": "error", "id": p_id})

print(json.dumps(results, separators=(',', ':')))
```

Problem 4: Detect Out-of-Stock

Description: The 'Detect Out-of-Stock' problem addresses the need to proactively identify products that have run out of stock. This involves iterating through the inventory data and flagging or listing all items whose current stock quantity is zero. This feature is vital for inventory reordering and preventing sales of unavailable items.

```
import sys
import json

input_data = sys.stdin.read()
if input_data.endswith('\n'):
    input_data = input_data[:-1]

lines = input_data.split('\n')

# Pre-loaded stock
stock = {"P1": 0, "P2": 5, "P3": 0, "P4": 10}
results = []

for line in lines:
    if line.strip() == "CHECK":
        empty_items = sorted(
            [item for item, count in stock.items() if count == 0]
        )
        results.append({"out_of_stock": empty_items})

print(json.dumps(results, separators=(',', ':')))
```

Problem 5: Export Inventory Summary

Description: The 'Export Inventory Summary' problem concerns the presentation of the entire inventory status in a clear and organized manner. This involves consolidating information about all products, such as their IDs, names, and current stock levels, and outputting it in a readily accessible format. This summary aids in overall inventory analysis and reporting.

```
import sys
import json

input_data = sys.stdin.read()
if input_data.endswith('\n'):
    input_data = input_data[:-1]

lines = input_data.split('\n')

# Starting Data
names = {"P1": "Apple", "P2": "Banana"}
stocks = {"P1": 5, "P2": 12}
results = []

for line in lines:
    if line.strip() == "REPORT":
        inventory = []

        for p_id in sorted(names.keys()):
            inventory.append({
                "id": p_id,
                "name": names[p_id],
                "stock": stocks[p_id]
            })

        results.append({"inventory": inventory})

print(json.dumps(results, separators=(',', ':')))
```

8. Learning Outcomes

• Gained practical experience in using Python dictionaries for data storage and retrieval.
• Developed proficiency in implementing basic CRUD (Create, Read, Update, Delete) operations for inventory management.
• Enhanced understanding of fundamental programming concepts like loops, conditionals, and functions.
• Learned to structure a simple command-line application.

9. Challenges Faced & Solutions

Challenge: Ensuring unique product IDs when adding new items.
Solution: Implemented logic to check if a product ID already exists in the dictionary before adding, prompting the user for a new ID or offering to update the existing entry.
Challenge: Handling invalid user input, such as non-numeric stock values or non-existent product IDs for updates/removals.
Solution: Incorporated try-except blocks to gracefully manage potential `ValueError` during type conversions and `KeyError` when accessing dictionary elements, providing informative error messages to the user.

10. Conclusion

Inventory Management Lite successfully demonstrates the application of core Python programming concepts to solve a practical problem. By leveraging dictionaries and basic logic, the system provides a functional and understandable approach to inventory tracking. This project serves as a solid foundation for understanding data management in software development and can be expanded upon to create more sophisticated inventory solutions.

11. Future Scope

• Implement persistent storage using files (e.g., CSV, JSON) to save and load inventory data between program runs.
• Add features for searching products by name or other attributes.
• Introduce user authentication and roles for more secure access control.

12. References

• Python Official Documentation: https://docs.python.org/3/
• Real Python: https://realpython.com/
• GeeksforGeeks - Python Tutorials: https://www.geeksforgeeks.org/python-programming-language/

13. Annexure

• Internship Completion Certificate
• Offer Letter